PROJECT LITERATURE REVIEW & TECHNICAL BENCHMARKS

Group 1: Core Benchmark & Foundation Papers

1. All-Spiking ECG Analysis for Arrhythmia Classification on Low-Power FPGA

Authors: M. A. Scrugli et al. (2026) | Journal: IEEE Sensors Journal

Link: 10.1109/JSEN.2026.3663715

This state-of-the-art paper achieves 98.4% accuracy on MIT-BIH using an SNN on a low-power Lattice FPGA. However, it relies on delta-modulation which requires real-time derivative calculation in hardware. Our project serves as a direct competitor, aiming to achieve comparable accuracy while eliminating the derivative pre-processing overhead by utilizing a much simpler LFSR-based rate encoder.

2. Surrogate Gradient Learning in Spiking Neural Networks

Authors: E. O. Neftci, H. Mostafa, F. Zenke (2019) | Journal: IEEE Signal Processing Magazine

Link: 10.1109/MSP.2019.2931595

This landmark paper introduces the mathematical framework to overcome the non-differentiable nature of binary spikes. By treating the spike as a smooth curve during the backward pass, it enables standard Backpropagation Through Time (BPTT). This exact methodology forms the theoretical basis for our Python training pipeline using the snnTorch library.

3. FPGA-Based Real-Time ECG Classification System Using Quantized Inception-ResNeXt

Authors: T. Cao et al. (2026) | Journal: IEEE TVLSI

Link: 10.1109/TVLSI.2024.3472270

This paper demonstrates the extreme upper limit of accuracy (99.5%) achievable on FPGAs using massive Convolutional Neural Networks with heavy DSP utilization. We cite this work to highlight our Pareto-tradeoff: CardioSpike-FPGA sacrifices a marginal fraction of accuracy in exchange for a massive reduction in logic gates and zero DSP dependency.

4. An Approximate Computing-Based SNN Neuron Model

Authors: H. Xia et al. (2026) | Journal: IEEE TCAS-I

Link: 10.1109/TCSI.2025.3624352

Explores approximate computing to remove DSP multipliers from neuromorphic hardware, proving that bit-shift substitutions boost energy efficiency by 6.75×. This provides strict mathematical and academic validation for our decision to set our membrane leak rate to $\beta = 0.9375$, which translates directly to a hardware right-shift (>> 4).

Group 2: SNN Hardware Architecture & Quantization

5. A Neuromorphic Processing System With Spike-Driven SNN Processor for Wearable ECG Classification

Authors: H. Chu et al. (2022) | Journal: IEEE TBioCAS

Link: IEEE Xplore

Details an event-driven SNN architecture for ECG that processes spikes asynchronously. Our project contrasts this by implementing a strictly synchronous, time-multiplexed Finite State Machine (FSM), which simplifies clock domain crossing and allows for easier deployment on standard commodity FPGAs without specialized asynchronous logic.

6. An Optimized Multi-Layer SNN Implementation in FPGA Without Multipliers

Authors: A. Mehrabi et al. (2023) | Journal: Elsevier Integration

Link: 10.1016/j.vlsi.2023.10.003

Demonstrates a highly efficient MAC-replacement strategy using adder trees and bit-shifts. Our VHDL architecture directly applies these principles to the ECG classification domain, validating our core VHDL `lif_neuron` module design which relies exclusively on subtractive accumulation.

7. QCSNN: A Memory-Efficient SNN for On-Device ECG Arrhythmia Detection

Authors: O. Banjo, B. Ghoraani (2025) | Journal: IEEE COINS

Link: IEEE Xplore

Focuses heavily on fixed-point quantization for SNN weights. This paper justifies our decision to quantize our trained floating-point weights into the Q1.7 fixed-point format, ensuring that all 2,384 weights fit entirely within a single 18Kb Block RAM tile without degrading classification performance.

8. Ultra-Energy-Efficient ECG Classification Processor with SNN Inference

Authors: R. Mao et al. (2022) | Journal: IEEE TBioCAS

Link: IEEE Xplore

Describes an ASIC implementation of an SNN for ECG that consumes less than a microjoule per inference. While our project targets reprogrammable FPGAs rather than custom ASICs, this work provides a comparative baseline for theoretical power limits when utilizing binary spike-based computing for cardiac monitoring.

9. Toward the Optimal Design and FPGA Implementation of SNNs

Authors: W. Guo et al. (2022) | Journal: IEEE TNNLS

Link: IEEE Xplore

Provides a comprehensive mathematical analysis of fixed-point precision requirements for LIF neurons. The authors conclude that Q8.8 arithmetic offers the optimal balance

between accuracy and hardware Look-Up Table (LUT) utilization, a guideline that we strictly adhered to in our VHDL datapath design.

10. A Generalized Hardware Architecture for Real-Time SNNs

Authors: D. Valencia et al. (2023) | Journal: Springer

Link: Springer Link

Outlines a generalized controller-datapath approach for executing deep networks on resource-constrained FPGAs. We adapted this methodology to create our `snn_top.vhd` Finite State Machine, allowing a single neuron circuit to sequentially process all 128 inputs to drastically save physical space on the chip.

Group 3: Data Preprocessing & Class Imbalance Handling

11. ECG Heartbeat Classification: A Deep Transferable Representation

Authors: M. Kachuee et al. (2018) | Journal: IEEE ICHI

Link: 10.1109/ICHI.2018.00092

This is the landmark paper that standardized the mapping of the 15+ complex PhysioNet MIT-BIH annotations into the 5 standard AAMI classes (N, S, V, F, Q). Our `preprocess.py` script utilizes their exact dataset extraction parameters (128-sample windows) to ensure our accuracy results are academically comparable to state-of-the-art literature.

12. Generalization of CNNs for ECG Classification Using GANs

Authors: A. M. Shaker et al. (2020) | Journal: IEEE Access

Link: 10.1109/ACCESS.2020.2987153

Highlights the severe class imbalance inherent in the MIT-BIH dataset, where Normal beats constitute over 80% of the samples. This paper justifies our critical need to implement the `augment_data.py` pipeline (Gaussian noise, time shifts) to synthetically boost minority classes and prevent the SNN from simply guessing "Normal" every time.

13. A Robust Deep CNN with Batch-Weighted Loss for Heartbeat Classification

Authors: A. Sellami, H. Hwang (2019) | Journal: Elsevier Expert Systems with Applications

Link: 10.1016/j.eswa.2018.12.027

Demonstrates that augmenting data alone is insufficient, and that the loss function itself must be penalized. We adopted their approach by utilizing inverse-frequency `class_weights` inside our PyTorch CrossEntropyLoss function, ensuring the network is heavily penalized for missing rare arrhythmias like Fusion beats.

14. Transfer Learning for ECG Classification

Authors: K. Weimann, T. O. F. Conrad (2021) | Journal: Scientific Reports

Link: 10.1038/s41598-021-84374-8

Provides the gold-standard methodology for baseline-wander removal and min-max normalization across diverse patient records. Our data pipeline applies their normalization

strategy, squashing all 128 samples into a precise [0, 1] range to ensure the LFSR Rate Encoder operates consistently across different patients.

Group 4: Rate Coding & Additional Support

15. SyncNN: Evaluating and Accelerating SNNs on FPGAs

Authors: S. Panchapakesan et al. (2020) | Journal: IEEE ICCAD

Link: IEEE Xplore

This paper provides a direct performance comparison between rate coding and temporal coding. We use their findings to defend our selection of rate coding over temporal coding, proving that for dense array time-series data like ECG, rate coding provides superior hardware simplicity at the edge.

16. A Low-Cost High-Speed Neuromorphic Hardware Based on SNN

Authors: E. Z. Farsa et al. (2019) | Journal: IEEE TCAS-II

Link: IEEE Xplore

A supplemental study reinforcing the viability of shift-and-add arithmetic in neuromorphic accelerators. We cite this to bulk our literature review and prove that our core arithmetic methodology is supported by multiple independent IEEE research groups.

17. Low-Energy and Fast SNN for Context-Dependent Learning on FPGA

Authors: H. Asgari et al. (2020) | Journal: IEEE TCAS-II

Link: IEEE Xplore

Validates the deployment of digital LIF accelerators specifically onto the Xilinx Spartan and Artix family of FPGAs. This relates directly to our project by assuring the feasibility of mapping our VHDL code onto low-tier Xilinx academic development boards.

18. Synaptic Activity and Hardware Footprint of SNNs in Digital Neuromorphic Systems

Authors: E. Lemaire et al. (2022) | Journal: ACM TECS

Link: ACM Digital Library

Analyzes the memory bottleneck caused by synaptic weight storage in large FPGA networks. Their research justifies our decision to keep the hidden layers constrained to 16 neurons each, ensuring all weights fit perfectly into a single internal BRAM tile rather than relying on slow external SDRAM.

19. Reconstruction of a Fully Paralleled Auditory SNN and FPGA Implementation

Authors: B. Deng et al. (2021) | Journal: IEEE TBioCAS

Link: IEEE Xplore

Although focused on auditory data, this paper deals with 1D time-series biological signals, similar to ECG. It validates the use of unrolled parallel inputs in the initial layer,

matching our architecture where 128 input pins are evaluated simultaneously by the rate encoder.

20. ECGTransForm: Adaptive ECG Arrhythmia Classification

Authors: H. El-Ghaish, E. Eldele (2024) | Journal: Elsevier

Link: 10.1016/j.compbiomed.2024.108210

A highly modern 2024 evaluation of adaptive classification methods on MIT-BIH. It provides the latest comparative metrics on how traditional machine learning techniques struggle with patient-specific morphology variations, underscoring the need for highly resilient neural networks like our SNN.